Method of Differentiation: Formula Sheet

First principle and notation

First principle:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Notation: $\frac{dy}{dx} = Dy = y_1 = y' = f'(x)$

Time derivative: $\dot{S} = \frac{dS}{dt}$

Standard derivatives

Power: $(x^n)' = nx^{n-1}$

Exponential: $(e^x)' = e^x,\ (a^x)' = a^x \ln a$

Log:

$$(\ln x)' = \frac{1}{x},\ (\log_a x)' = \frac{1}{x \ln a}$$

$$(\sin x)' = \cos x,\ (\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x,\ (\cot x)' = -\operatorname{cosec}^2 x$$

$$(\sec x)' = \sec x \tan x,\ (\operatorname{cosec} x)' = -\operatorname{cosec} x \cot x$$

$$(\sin^{-1} x)' = -(\cos^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\tan^{-1} x)' = -(\cot^{-1} x)' = \frac{1}{1+x^2}$$

$$(\sec^{-1} x)' = -(\operatorname{cosec}^{-1} x)' = \frac{1}{|x|\sqrt{x^2-1}}$$

Rules

Product: $(fg)' = f'g + fg'$

Three factors:

$$(fgh)' = \frac{1}{2}[(fg)'h + (gh)'f + (hf)'g]$$

Quotient:

$$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$

Chain:

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

Reciprocal:

$$\frac{dy}{dx} = 1 \Big/ \frac{dx}{dy}$$

Techniques

Variable power:

$$(f^g)' = f^g \left[g' \ln f + \frac{gf'}{f}\right]$$

Log differentiation: $\frac{y'}{y} = \frac{d}{dx} \ln y$

Parametric:

$$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}}$$

f w.r.t. g:

$$\frac{df}{dg} = \frac{f'(x)}{g'(x)}$$

Implicit: $\frac{dy}{dx} = -\frac{f_x}{f_y}$

Homogeneous $= 0$: $y' = \frac{y}{x},\ y'' = 0$

Inverse: $g'(b) = \frac{1}{f'(a)},\ f(a) = b$

Inverse, second: $g''(b) = -\frac{f''(a)}{(f'(a))^3}$

Higher order derivatives

Second derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Parametric:

$$\frac{d^2y}{dx^2} = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^3}$$

Swap variables:

$$\frac{d^2x}{dy^2} = -\frac{y_2}{y_1^3}$$

$y = \sin^{-1} x$: $(1-x^2)y_2 - xy_1 = 0$

ITF analysis

$\sin^{-1} \frac{2x}{1+x^2}$: $2\tan^{-1} x\ (|x| \le 1)$, $\pm\pi - 2\tan^{-1} x\ (x \gtrless \pm 1)$

$$\cos^{-1} \frac{1-x^2}{1+x^2} = 2|\tan^{-1} x|$$

$\tan^{-1} \frac{2x}{1-x^2}$: $2\tan^{-1} x\ (|x| < 1)$, $\pi + 2\tan^{-1} x\ (x < -1)$, $2\tan^{-1} x - \pi\ (x > 1)$

Miscellaneous and determinants

Coefficient: $a_r = \frac{f^{(r)}(0)}{r!}$

Divisibility: $(x-a)^2 \mid f \iff f(a) = f'(a) = 0$

Binomial sum: $\sum r\binom{n}{r} = n2^{n-1}$

Determinant: $\Delta' = \sum$ (one row differentiated)

L'Hospital and integrals

L'Hospital: $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ for $\frac{0}{0}, \frac{\infty}{\infty}$

$0^0, \infty^0$: $\lim f^g = e^{\lim g \ln f}$

Indefinite: $\frac{d}{dx} \int f\,dx = f$

Leibniz:

$$\frac{d}{dx} \int_g^h f(t)dt = f(h)h' - f(g)g'$$